Chapter 2

Electrostatics

Problem 2.1

(a) Twelve equal charges, q, are situated at the corners of a regular 12-sided polygon (for instance, one on each numeral of a clock face). What is the net force on a test charge Q at the center?

(b) Suppose on of your 12 qs is removed (the one at "6 o'clock"). What is the force on Q? Explain your reasoning carefully.

(c) Now 13 equal charges q, are placed at the corners of a regular 13-sided polygon. What is the force on a test charge Q at the center?

(d) If one of the 13 qs is removed, what is the force on Q? Explain your reasoning.

Solution:

(a) Due to the symmetry, the net force on a test charge at the center is zero. This is because each opposite charge q on the 12-sided polygon cancel each others force at the center. The force from the charge at 1 o'clock cancels the force from the charge at 7 o'clock, etc.

(b) If the charge at 6 o'clock is removed, the net force on a test charge at the center is

$$\frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}$$

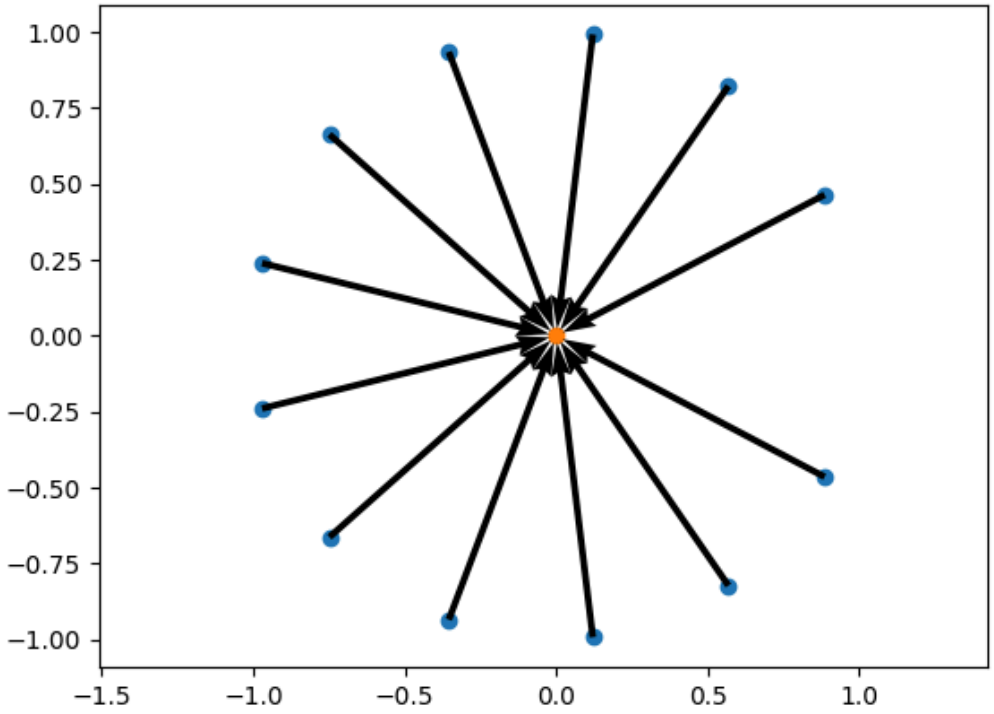
directed straight down (straight up) if the charge q and the test charge Q are both positive or both negative (if they are positive/negative or negative/positive respectively).

(c) Again, due to symmetry, the net force at the center is zero.

(d) The force on the center would again be

$$\frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}$$

directed towards the removed charge (or opposite direction, if the charges q and the test charge Q are of opposite signs). This can either be explained by simply adding a charge of opposite sign at the removed charge, or by the same idea as in (b), from the following image



All vertical components of the forces cancel, giving the total force as the sum of the individual charges:

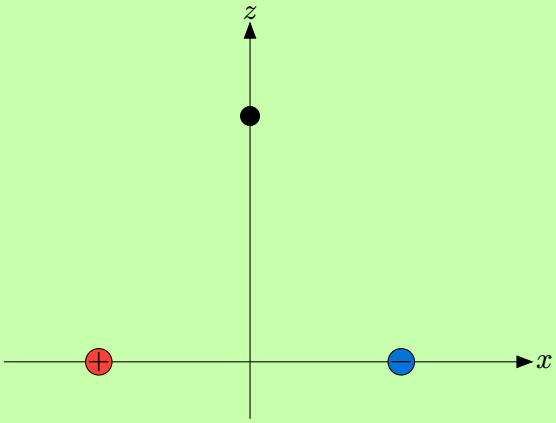
$$\frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}\sum_{k=1}^{13}\cos\left(\frac{2\pi k}{13}\right) = \frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}\hat{\mathbf{r}} \text{ with } \hat{\mathbf{r}} \text{ pointing towards the right}$$

□

Problem 2.2

Find the electric field (magnitude and direction) a distance z above the mid-point between equal and opposite charges ($\pm q$), a distance d apart.

Solution:



The electric field from a point charge is given by

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}}$$

where r is the distance between the charge and the field point and $\hat{\boldsymbol{r}}$ is the unit separation vector from the charge to the field point. The field at our measurement point will have no vertical (z-directed) component since the positive charge will create a field in the positive z-direction with the same magnitude that the negative charge will create a field in the negative z-direction. Therefore the total field will be directed towards $+x$ with a magnitude twice that of what the positive charge creates. The separation vector magnitude becomes

$$r = \sqrt{z^2 + \left(\frac{d}{2}\right)^2}$$

and

$$\boldsymbol{r} = \frac{d}{2}\hat{\mathbf{x}}$$
$$\downarrow$$
$$\hat{\boldsymbol{r}} = \frac{\boldsymbol{r}}{r} = \frac{d}{2\,r}\hat{\mathbf{x}}$$

resulting in the total electric field from both charges

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{qd}{\sqrt{z^2 + \left(\frac{d}{2}\right)^2}^3}\hat{\mathbf{x}}$$

□

Problem 2.3

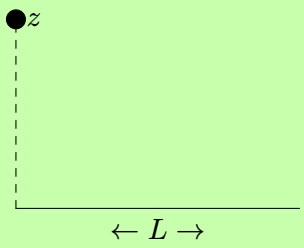
Find the electric field a distance z above one end of a straight line segment of length L that carries a uniform line charge λ. Check that your formula in consistent with what you would expect for the case $z \gg L$.

Solution:

The electric field from a continuous line charge distribution is given by

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\int \frac{\lambda(\mathbf{r}')}{\boldsymbol{\mathcal{r}}^2}\,\mathrm{d}l'$$

The setup is shown in the figure below



The separation vector $\boldsymbol{\mathcal{r}}$ becomes

$$-x\hat{\mathbf{x}} + z\hat{\mathbf{z}}$$

which gives

$$\hat{\boldsymbol{\mathcal{r}}} = \frac{-x\hat{\mathbf{x}} + z\hat{\mathbf{z}}}{\sqrt{x^2+z^2}}$$

This means our integral becomes, since λ is constant,

$$\mathbf{E} = \frac{\lambda}{4\pi\epsilon_0}\int_0^L \frac{-x\hat{\mathbf{x}} + z\hat{\mathbf{z}}}{\sqrt{x^2+z^2}^3}\,\mathrm{d}x$$

The following standard integrals will be used in the calculations:

$$\int \frac{1}{\sqrt{x^2+a^2}^3}\,\mathrm{d}x = \frac{1}{a^2}\frac{x}{\sqrt{x^2+a^2}}$$

$$\int \frac{x}{\sqrt{x^2+a^2}}\,\mathrm{d}x = -\frac{1}{\sqrt{x^2+a^2}}$$

Starting with the x-component gives

$$E_x = -\frac{\lambda}{4\pi\epsilon_0}\int_0^L \frac{x}{\sqrt{x^2+z^2}^3}\,\mathrm{d}x = -\frac{\lambda}{4\pi\epsilon_0}\left[-\frac{1}{\sqrt{x^2+z^2}}\right]_0^L =$$

$$= \frac{\lambda}{4\pi\epsilon_0}\left(\frac{1}{z} - \frac{1}{\sqrt{L^2+z^2}}\right)$$

And the z-component

$$E_z = \frac{\lambda}{4\pi\epsilon_0}\int_0^L \frac{z}{\sqrt{x^2+z^2}^3}\,\mathrm{d}x = \frac{\lambda z}{4\pi\epsilon_0}\left[\frac{1}{z^2}\frac{x}{\sqrt{x^2+z^2}}\right]_0^L = \\ = \frac{\lambda}{4\pi\epsilon_0 z}\left(\frac{L}{\sqrt{L^2+z^2}}\right)$$

The electric field is then

$$\mathbf{E} = \frac{\lambda}{4\pi\epsilon_0}\left(\frac{1}{z}-\frac{1}{\sqrt{L^2+z^2}}\right)\hat{\mathbf{x}} + \frac{\lambda}{4\pi\epsilon_0 z}\left(\frac{L}{\sqrt{L^2+z^2}}\right)\hat{\mathbf{z}}$$

In the case that $z \gg L$, we get

$$\mathbf{E} = \frac{\lambda}{4\pi\epsilon_0}\left(\frac{1}{z}-\frac{1}{z}\right)\hat{\mathbf{x}} + \frac{\lambda}{4\pi\epsilon_0 z}\left(\frac{L}{z}\right)\hat{\mathbf{z}} = \frac{\lambda L}{4\pi\epsilon_0 z^2}\hat{\mathbf{z}}$$

which is of the same form as the field at a point z from a point charge of magnitude λL. □

Problem 2.4

Find the electric field a distance z above the center of a square loop (side length d) carrying a uniform line charge λ.

Solution:

Due to the symmetry, it is enough to calculate the contribution from one of the sides, and multiplying this result by 4. Only the z-component will be calculated since the x- and y-components will cancel.

Starting with determining the separation vector for one of the sides ($x : -\frac{a}{2} \to \frac{a}{2}, y = \frac{a}{2}$)

$$\boldsymbol{\mathcal{r}} = x\hat{\mathbf{x}} + \frac{a}{2}\hat{\mathbf{y}} + z\hat{\mathbf{z}} = z\hat{\mathbf{z}}$$

and the magnitude

$$\mathcal{r} = \sqrt{x^2 + \left(\frac{a}{2}\right)^2 + z^2}$$

gives

$$\hat{\boldsymbol{\mathcal{r}}} = \frac{z\hat{\mathbf{z}}}{\sqrt{x^2+\left(\frac{a}{2}\right)^2+z^2}}$$

Using these, the electric field is given by

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\int_{-\frac{a}{2}}^{\frac{a}{2}} \frac{\lambda z\hat{\mathbf{z}}}{\sqrt{x^2+\left(\frac{a}{2}\right)^2+z^2}^3}\,\mathrm{d}x$$

$$E_x = \frac{\lambda z}{4\pi\epsilon_0}\int_{-\frac{a}{2}}^{\frac{a}{2}} \frac{1}{\sqrt{x^2+\sqrt{\left(\frac{a}{2}\right)^2+z^2}^2}^3}\,\mathrm{d}x = \frac{\lambda z}{4\pi\epsilon_0}\left[\frac{x}{\left(\left(\frac{a}{2}\right)^2+z^2\right)\sqrt{x^2+\left(\frac{a}{2}\right)^2+z^2}}\right]_{-\frac{a}{2}}^{\frac{a}{2}} =$$

$$= \frac{\lambda z}{4\pi\epsilon_0}\frac{a}{\left(\left[\frac{a^2}{4}+z^2\right]\sqrt{\frac{a^2}{2}+z^2}\right)}$$

Multiplying this by four gives the final result

$$\mathbf{E} = \frac{\lambda z a}{\pi\epsilon_0\left(\frac{a^2}{4}+z^2\right)\sqrt{\frac{a^2}{2}+z^2}}\hat{\mathbf{z}}$$

□

Problem 2.5

Find the electric field a distance z above the center of a circular loop of radius r that carries a uniform line charge λ.

Solution:

Using cylindrical coordinates, the separation vector $\boldsymbol{\mathscr{r}}$ is

$$\boldsymbol{\mathscr{r}} = r\hat{\mathbf{r}} + z\hat{\mathbf{z}} = [\text{symmetry}] = z\hat{\mathbf{z}}$$

and the magnitude

$$\mathscr{r} = \sqrt{r^2+z^2}$$

gives

$$\hat{\boldsymbol{\mathscr{r}}} = \frac{z}{\sqrt{r^2+z^2}}\hat{\mathbf{z}}$$

The circular loop at radius r has the integral form $\int_0^{2\pi} r\,\mathrm{d}\theta$. The integral that gives the resulting electric field is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\int_0^{2\pi} \frac{\lambda z\hat{\mathbf{z}}}{\sqrt{r^2+z^2}^3} r\,\mathrm{d}\theta = \frac{\lambda z r}{2\epsilon_0\sqrt{r^2+z^2}^3}\hat{\mathbf{z}}$$

□

Problem 2.6

Find the electric field a distance z above the center of a flat circular disk of radius R that carries a uniform surface charge σ. What does your formula give in the limit $R \to \infty$? Also check the case $z \gg R$.

Solution:

The electric field for a surface charge is given by

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int_{\mathcal{S}} \frac{\sigma(\mathbf{r}')}{r\imath^2}\hat{\boldsymbol{\imath}}\,\mathrm{d}a'$$

Our disk is given by the cylindrical coordinates s and ϕ

$$s : 0 \to R, \quad \phi : 0 \to 2\pi$$

The integration is

$$\mathrm{d}a' = s\,\mathrm{d}s\,\mathrm{d}\phi$$

The separation vector is

$$\boldsymbol{\imath} = z\hat{\mathbf{z}} - s\hat{\mathbf{s}}$$
$$\downarrow$$
$$\imath = \sqrt{z^2 + s^2}$$
$$\downarrow$$
$$\hat{\boldsymbol{\imath}} = \frac{z\hat{\mathbf{z}} - s\hat{\mathbf{s}}}{\sqrt{z^2 + s^2}}$$

Due to the symmetry of the problem, the s-dependence will add up to zero, so we will ignore the s-component in the unit separation vector. The electric field at the z-axis is then

$$\mathbf{E}(z) = \frac{1}{4\pi\epsilon_0}\int_0^R\int_0^{2\pi} \frac{\sigma z\hat{\mathbf{z}}}{\sqrt{z^2+s^2}^3}s\,\mathrm{d}s\,\mathrm{d}\phi =$$

$$= \frac{2\pi z\sigma\hat{\mathbf{z}}}{4\pi\epsilon_0}\int_0^R \frac{s}{\sqrt{z^2+s^2}^3}\,\mathrm{d}s = \frac{2\pi z\sigma\hat{\mathbf{z}}}{4\pi\epsilon_0}\left[-\frac{1}{\sqrt{z^2+s^2}}\right]_0^R =$$

$$= \frac{z\sigma}{2\epsilon_0}\left[\frac{1}{z} - \frac{1}{\sqrt{z^2+R^2}}\right]\hat{\mathbf{z}}$$

In the limit that $R \to \infty$, the electric field is given by

$$\frac{z\sigma}{2\epsilon_0}\left[\frac{1}{z} - \frac{1}{R}\right]\hat{\mathbf{z}} = \frac{\sigma}{2\epsilon_0}\hat{\mathbf{z}}$$

When $z \gg R$, naïvely setting the square root term to $\frac{1}{\sqrt{z^2+R^2}} = \frac{1}{\sqrt{z^2}} = \frac{1}{z}$ would give the (somewhat) incorrect result zero. The square root expression needs to be Taylor expanded with the rule (assuming $|x| \ll 1$)

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + ...$$

Rewriting the square root term

$$\frac{1}{\sqrt{z^2+R^2}} = \frac{1}{z}\frac{1}{\sqrt{1+\frac{R^2}{z^2}}} = \frac{1}{z}\left(1+\frac{R^2}{z^2}\right)^{-\frac{1}{2}}$$

$$\downarrow$$

$$\frac{1}{\sqrt{z^2+R^2}} \approx \frac{1}{z}\left(1-\frac{1}{2}\frac{R^2}{z^2}\right)$$

which gives the electric field

$$\frac{z\sigma}{2\epsilon_0}\left[\frac{1}{z} - \frac{1}{z}\left(1-\frac{1}{2}\frac{R^2}{z^2}\right)\right]\hat{\mathbf{z}} = \frac{z\sigma}{2\epsilon_0}\frac{1}{2}\frac{R^2}{z^3}\hat{\mathbf{z}}$$

Expressing the charge density and the radius in terms of the total charg Q would give $Q = \sigma\pi R^2$ which results in the electric field

$$\mathbf{E}(z) = \frac{Q}{4\pi\epsilon_0 z^2}\hat{\mathbf{z}}$$

□

Problem 2.7

Problem 2.8

Problem 2.9

Suppose the electric field in some region is found to be $\mathbf{E} = kr^3\mathbf{r}$, in spherical coordinates (k is some constant).

(a) Find the charge density ρ.

(b) Find the total charge contained in a sphere of radius R, centered at the origin. (Do it two different ways.)

Solution:

(a) Gauss's law in differential form is

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \frac{1}{\epsilon_0}\rho \rightarrow \rho = \epsilon_0 \boldsymbol{\nabla} \cdot \mathbf{E}$$

The divergence of $\mathbf{E}$ is

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \boldsymbol{\nabla} \cdot kr^3\hat{\mathbf{r}} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 kr^3) = 5kr^2$$

so the charge density ρ is

$$\rho = \epsilon_0 5kr^2$$

(b) The total charge Q can be obtained in two ways; integrating the charge density over the volume, or integrating the electric field over the bounding surface.

Integrating the charge density over the volume of a sphere with radius R gives

$$Q = \int_{\mathcal{V}} \epsilon_0 5kr^2 r^2 \sin\theta \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}\phi$$

$$\downarrow$$

$$5k\epsilon_0 \int_0^R r^4 \left(\int_0^\pi \sin\theta \left(\int_0^{2\pi} \mathrm{d}\phi \right) \mathrm{d}\theta \right) \mathrm{d}r =$$

$$= 20k\pi\epsilon_0 \int_0^R r^4 \, \mathrm{d}r = \boxed{Q = 4\pi\epsilon_0 kR^5}$$

Integrating the electric field over the bounding surface of the sphere with radius R gives

$$\frac{Q}{\epsilon_0} = \int_{\mathcal{S}} kr^3 r^2 \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi$$

$$\downarrow$$

$$Q = \epsilon_0 kR^5 \int_0^\pi \sin\theta \left(\int_0^{2\pi} \mathrm{d}\phi \right) \mathrm{d}\theta \rightarrow \boxed{Q = 4\pi\epsilon_0 kR^5}$$

□

Problem 2.10

A charge q sits at the back corner of a cube, as shown in the figure. What is the flux of $\mathbf{E}$ through the shaded side?

Solution:

Adding 7 cubes to totally enclose the charge gives a total of 24 faces that the flux is passing through. This means that the flux of this one surface is 1/24 of the total flux given by the charge q.

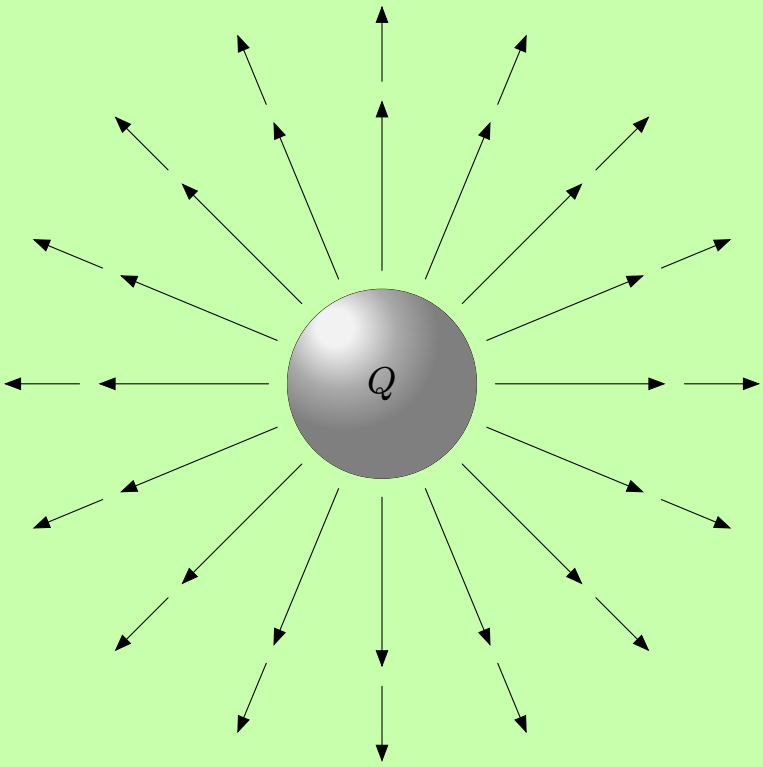
$$[\text{Flux through the shaded surface}] = \frac{q}{24\epsilon_0}$$

□

Problem 2.11

Draw a field-line diagram for a uniformly charged solid sphere.

Solution:



□

Problem 2.12

Use Gauss's law to find the electric field inside and outside a spherical shell of radius R that carries a uniform surface charge density σ.

Solution:

Gauss's law states that the closed surface integral of the electric field is equal to the charge enclosed in that surface.

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

Using the Gaussian surface of a sphere, the charge enclosed inside the spherical shell is zero, and outside it is

$$Q_{\text{enc}} = 4\pi R^2 \sigma$$

The electric field is always directed towards $\hat{\mathbf{r}}$ and $\mathrm{d}\mathbf{a} = r^2 \sin\theta \, \mathrm{d}\theta \mathrm{d}\phi$ gives

$$Er^2 4\pi = \frac{4\pi R^2 \sigma}{\epsilon_0}$$

$$\downarrow$$

$$\boxed{\mathbf{E} = \frac{R^2 \sigma}{\epsilon_0 r^2} \hat{\mathbf{r}}, \quad r > R}$$

where R is the radius of the spherical shell and r is the distance from the center □

Problem 2.13

Use Gauss's law to find the electric field inside a uniformly charged solid sphere (charge density ρ).

Solution:

For our Gaussian surface, we will use the surface of a sphere. Gauss's law is

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

While our Gaussian surface is inside the uniformly charged sphere, the charged enclosed is the volume integral of the charge density, where the volume is a sphere. Let a be the radius of our Gaussian surface.

$$Ea^2 \int_0^{\pi} \int_0^{2\pi} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi = \frac{1}{\epsilon_0} \int_0^{a} \int_0^{\pi} \int_0^{2\pi} \rho r^2 \sin\theta \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}\phi$$

$$\downarrow$$

$$Ea^2 4\pi = \frac{\rho a^3 4\pi}{3\epsilon_0}$$

$$\downarrow$$

$$\mathbf{E} = \frac{\rho a}{3\epsilon_0} \hat{\mathbf{r}}$$

□

Problem 2.14

Find the electric field a distance s from an infinitely long straight wire that carries a uniform line charge λ.

Solution:

The Gaussian surface for this problem is a cylinder. The enclosed charge of this cylinder is

$$Q_{\text{enc}} = \lambda z$$

where z is the height of the cylinder. The top and bottom disks of the cylinder are not accounted for since the field there is zero due to symmetry. Gauss's law becomes

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{a} = \frac{\lambda z}{\epsilon_0}$$

$$\downarrow$$

$$\int_0^{z} \int_0^{2\pi} Es \, \mathrm{d}\phi \, \mathrm{d}\theta = \frac{\lambda z}{\epsilon_0}$$

$$\downarrow$$

$$Esz2\pi = \frac{\lambda z}{\epsilon_0}$$

$$\downarrow$$

$$E = \frac{\lambda}{2\pi s \epsilon_0}$$

$$\downarrow$$

$$\mathbf{E} = \frac{\lambda}{2\pi s \epsilon_0} \hat{\mathbf{s}}$$

□

Problem 2.15

Find the electric field inside a sphere that carries a charge density proportional to the distance from the center, $\rho = kr$, for some constant k.

Solution:

The Gaussian surface for this problem is the surface of a sphere. The enclosed charge is obtained by integrating the charge density. Let r be the radius of our Gaussian surface, that is, $r : 0 \to R$, where R is the radius of the charged sphere. Gauss's law becomes

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

$$\downarrow$$

$$\int_0^{\pi} \int_0^{2\pi} Er^2 \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi = \frac{1}{\epsilon_0} \int_0^{r} \int_0^{\pi} \int_0^{2\pi} kr^3 \sin\theta \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}\phi$$

$$\downarrow$$

$$Er^2 4\pi = \frac{kr^4 4\pi}{4\epsilon_0}$$

$$\downarrow$$

$$E = \frac{kr^2}{4\epsilon_0}$$

$$\downarrow$$

$$\boxed{\mathbf{E} = \frac{kr^2}{4\epsilon_0} \hat{\mathbf{r}}}$$

□

Problem 2.16

A thick spherical shell carries charge density

$$\rho = \frac{k}{r^2}, \quad (a \leq r \leq b)$$

Find the electric field in the following three regions:

(i) $r < a$, (ii) $a < r < b$, (iii) $r > b$

Plot $|\mathbf{E}|$ as a function of r, for the case $b = 2a$.

Solution:

Inside region (i), there is no enclosed charge, therefore the electric field is zero in this region. In region (ii), we construct a spherical Gaussian surface, and integrate the charge density from a to r ($a < r < b$)

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

$$\downarrow$$

$$\int_0^{\pi} \int_0^{2\pi} E r^2 \sin\theta \, \mathrm{d}\theta \mathrm{d}\phi = \frac{1}{\epsilon_0} \int_a^r \int_0^{\pi} \int_0^{2\pi} \frac{k}{r^2} r^2 \sin\theta \, \mathrm{d}r \mathrm{d}\theta \mathrm{d}\phi$$

$$\downarrow$$

$$E r^2 4\pi = \frac{k 4\pi}{\epsilon_0} (r - a)$$

$$\downarrow$$

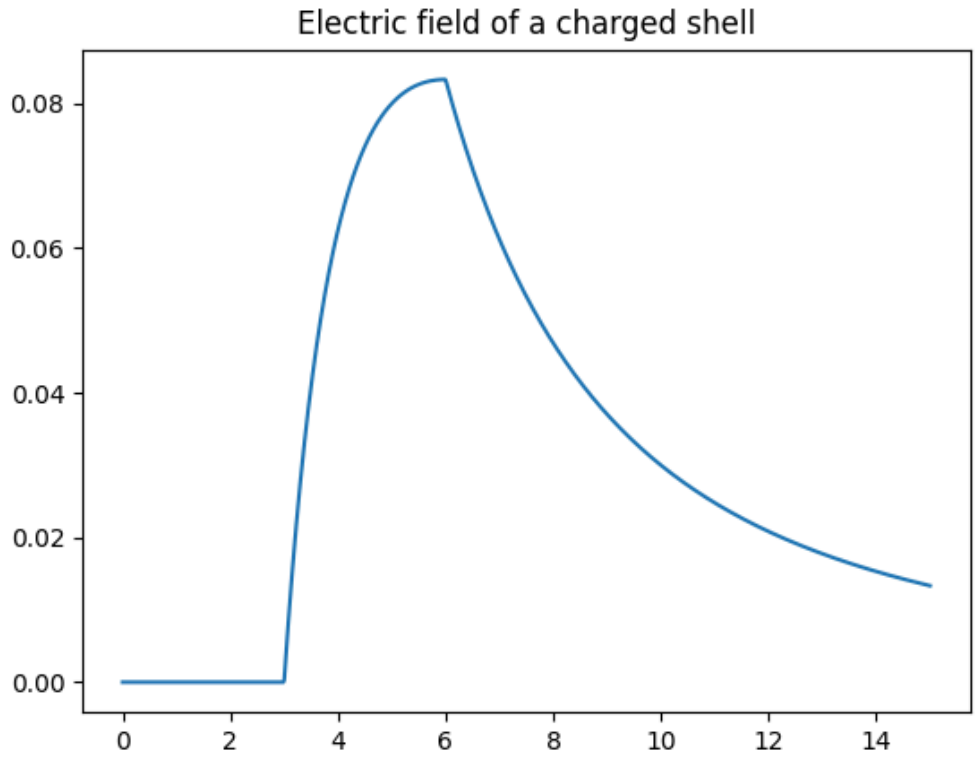
$$E = \frac{k(r-a)}{r^2 \epsilon_0}$$

$$\downarrow$$

$$\mathbf{E} = \frac{k(r-a)}{r^2 \epsilon_0} \hat{\mathbf{r}}$$

In region (iii), the total charge enclosed is obtained by integrating the charge density. We get the same result as in (ii) but now $(r - a) \to (b - a)$ since the integral is taken from a to b.

$$\mathbf{E} = \frac{k(b-a)}{r^2 \epsilon_0}, \quad r > b$$



□

Problem 2.17

A long coaxial cable carries a uniform volume charge density ρ on the inner cylinder (radius a), and a uniform surface charge density on the outer cylindrical shell (radius b). This surface charge is negative and is of just the right magnitude that the cable as a whole is electrically neutral. Find the electric field in each of the three regions:

(i) inside the inner cylinder ($s < a$),

(ii) between the cylinders ($a < s < b$),

(iii) outside the cable ($s > b$).

Plot $|\mathbf{E}|$ as a function of s.

Solution:

Using a cylindrical Gaussian surface gives for region (i)

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{a} = Q_{\text{enc}}$$

$$\downarrow$$

$$E 2\pi s z = \frac{1}{\epsilon_0} \int_0^a \rho s \, \mathrm{d}s \, \mathrm{d}\phi \, \mathrm{d}z$$

$$\downarrow$$

$$E 2\pi s z = \frac{\rho s^2 2\pi z}{2\epsilon_0}$$

$$\downarrow$$

$$E = \frac{\rho s}{2\epsilon_0}$$

$$\downarrow$$

$$\mathbf{E} = \frac{\rho s}{2\epsilon_0} \hat{\mathbf{s}}, \quad s < a$$

In region (ii), the total enclosed charge is the charge density multiplied by the volume, which is

$$Q_{\text{enc}} = \rho \pi a^2 z$$

which gives the electric field

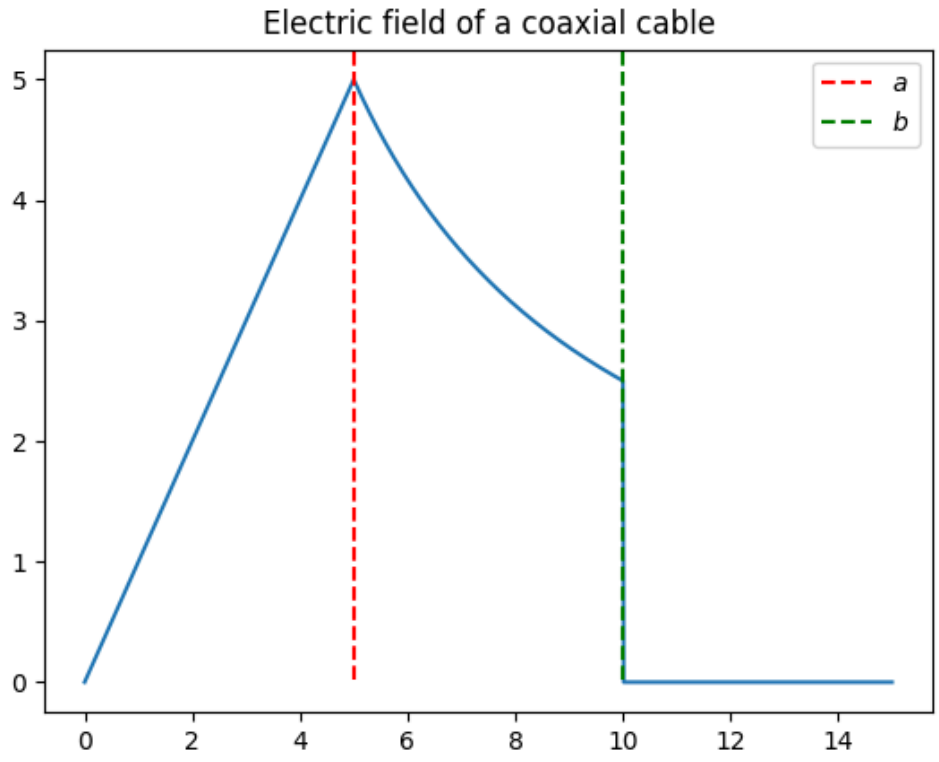
$$E 2\pi s z = \frac{\rho \pi a^2 z}{\epsilon_0}$$

$$\downarrow$$

$$\mathbf{E} = \frac{\rho a^2}{2\epsilon_0 s} \hat{\mathbf{s}}, \quad a < s < b$$

In region (iii), the total charge is zero since the positive and negative parts cancel,

$$\mathbf{E} = 0, \quad s > b$$



□

Problem 2.18

An infinite plane slab, of thickness $2d$, carries a uniform volume charge density ρ (FIGURE). Find the electric field, as a function of y, where $y = 0$ at the center. Plot E versus y, calling E positive when it points in the $+y$ direction and negative when it points in the $-y$ direction.

Solution:

We create a Gaussian surface as a box. The enclosed charge is simply the box area A times the position y

$$\oint_S \mathbf{E} \cdot \mathrm{d}\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

$$2EA = \frac{1}{\epsilon_0} \rho 2dA$$

$$\mathbf{E} = \frac{\rho d}{\epsilon_0}\hat{\mathbf{y}}, \qquad y > 0$$

$$\mathbf{E} = \frac{\rho d}{\epsilon_0}(-\hat{\mathbf{y}}), \quad y < 0$$

Inside:

$$2EA = \frac{1}{\epsilon_0}\rho 2yA$$

$$\mathbf{E} = \frac{\rho y}{\epsilon_0}\hat{\mathbf{y}}$$

□

Problem 2.19

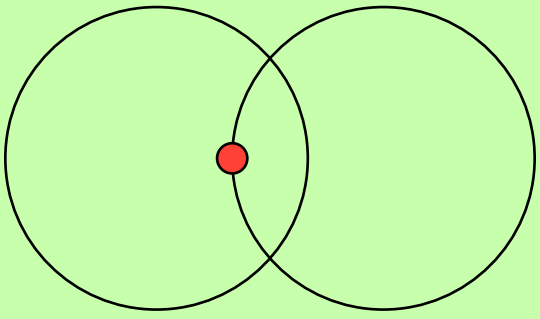
Two spheres, each of radius R and carrying uniform volume charge densities $+\rho$ and $-\rho$, respectively, are placed so that they partially overlap (FIGURE). Call the vector from the positive center to the negative center d. Show that the field in the region of overlap is constant, and find its value. [Hint: Use the answer to Prob. 2.13.]

Solution:

The result of Prob. 2.13 is

$$\mathbf{E} = \frac{\rho a}{3\epsilon_0}\hat{\mathbf{r}}$$

Aligning the spheres in the $\hat{\mathbf{x}}$-direction and using superposition on the red point below:



Here, the field due to the positive sphere is

$$\mathbf{E}_+ = \frac{\rho(d-R)}{3\epsilon_0}\hat{\mathbf{x}}$$

and the field due to the negative sphere is

$$\mathbf{E}_{-} = \frac{\rho R}{3\epsilon_0}\hat{\mathbf{x}}$$

Since the field is constant in the overlapping region, it will be equal to the sum of these two contributions at all points in the overlap.

Adding them together gives

$$\boxed{\mathbf{E} = \frac{\rho d}{3\epsilon_0}\hat{\mathbf{x}}}$$

Generalizing in any direction simply gives

$$\boxed{\mathbf{E} = \frac{\rho d}{3\epsilon_0}\hat{\mathbf{d}}}$$

where $\hat{\mathbf{d}}$ is the vector pointing from the center of the positively charged sphere, towards the center of the negatively charged sphere. ! :) □

Problem 2.20

Problem 2.21

Problem 2.22

Find the potential inside and outside a uniformly charged solid sphere whose radius is R and whose total charge is q. Use infinity as your reference point. Compute the gradient of V in each region, and check that it yields the correct field. Sketch $V(\mathbf{r})$.

Solution:

Outside the sphere, where $r > R$:

$$V_{\text{o}} = -\frac{1}{4\pi\epsilon_0}\int_{\infty}^{r}\frac{q}{r^2}\,\mathrm{d}r = \frac{q}{4\pi\epsilon_0 r}$$

Inside the sphere, we have the result above at $r = R$ minus the integral from the sphere surface with $r < R$. The total charge is the ratio between the full sphere with radius R and the sphere with radius r: $q = q\frac{r^3}{R^3}$

$$V_{\mathrm{i}} = \frac{q}{4\pi\epsilon_0 R} - \frac{1}{4\pi\epsilon_0}\int_R^r \frac{q}{r^2}\,\mathrm{d}r =$$

$$\frac{q}{4\pi\epsilon_0 R} - \frac{1}{4\pi\epsilon_0}\int_R^r \frac{qr}{R^3}\,\mathrm{d}r = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{R} - \left[\frac{r^2}{2R^3}\right]_R^r\right) =$$

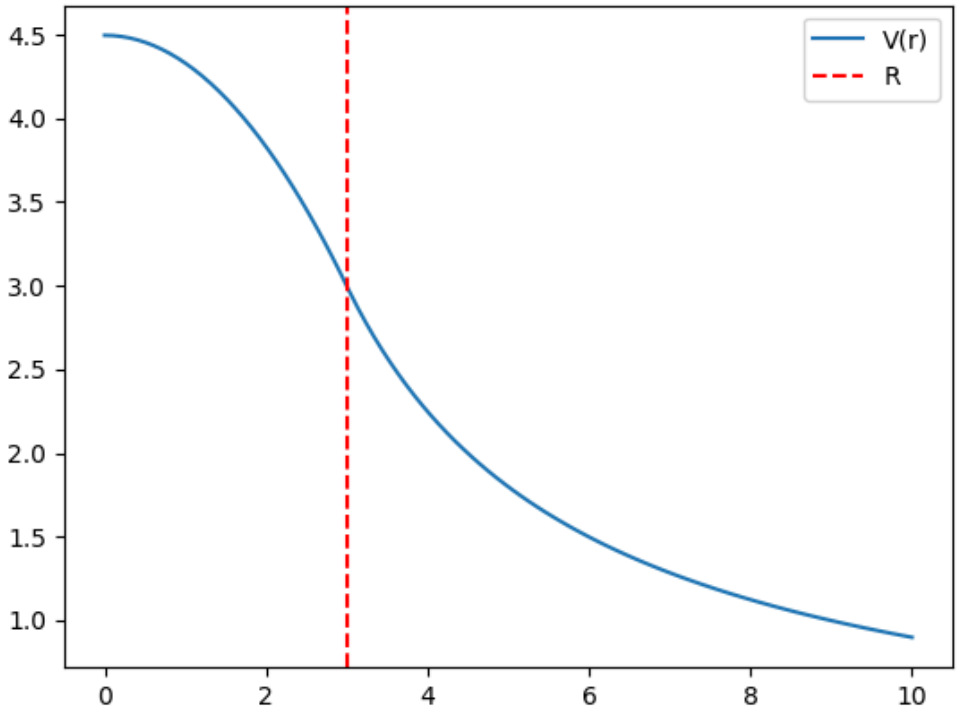
$$= \frac{q}{4\pi\epsilon_0}\left(\frac{1}{R} - \frac{1}{R^3}\left(\frac{r^2 - R^2}{2}\right)\right)$$

The gradient of the potential outside the sphere gives the electric field outside:

$$\mathbf{E}_{\mathrm{o}} = -\boldsymbol{\nabla} V = \frac{q}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}}$$

and inside gives the electric field

$$\mathbf{E}_{\mathrm{i}} = -\boldsymbol{\nabla} V = \frac{qr}{4\pi\epsilon_0 R^3}\hat{\mathbf{r}}$$



□

Problem 2.23

Find the potential a distance s from an infinitely long straight wire that carries a uniform line charge λ. Compute the gradient of your potential, and check that it yields the correct field.

Solution:

From Gauss' Law, the field is

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s}\hat{\mathbf{s}}$$

The potential is then

$$V(\mathbf{s}) = -\int_a^s \frac{\lambda}{2\pi\epsilon_0 s}\,\mathrm{d}s = \frac{\lambda}{2\pi\epsilon_0}[\ln(s)]_s^a =$$
$$= \frac{\lambda}{2\pi\epsilon_0}\ln\left(\frac{a}{s}\right)$$

The gradient of this potential, the electric field, is simply

$$\mathbf{E} = -\boldsymbol{\nabla}V = -\frac{\lambda}{2\pi\epsilon_0 s}\hat{\mathbf{s}}$$

□

Problem 2.24

For the charge configuration of Prob. 2.15, find the potential at the center, using infinity as your reference point.

Solution:

The electric field outside is

$$\mathbf{E} = \frac{k(b-a)}{r^2\epsilon_0}, \quad r > b$$

and in the shell

$$\mathbf{E} = \frac{k(r-a)}{r^2\epsilon_0}\hat{\mathbf{r}}$$

and zero inside.

The potential from ∞ to b is

$$V_{\infty\to b} = -\int_{\infty}^{b} \frac{k(b-a)}{r^2\epsilon_0}\,\mathrm{d}r = \left[\frac{k(b-a)}{r\epsilon_0}\right]_{\infty}^{b} = \frac{k(b-a)}{b\epsilon_0} = \frac{k}{\epsilon_0} - \frac{ka}{b\epsilon_0}$$

The potential from b to a is

$$V_{b\to a} = -\int_{b}^{a} \frac{k(r-a)}{r^2\epsilon_0}\,\mathrm{d}r = \int_{b}^{a} \frac{ka}{r^2\epsilon_0} - \frac{k}{r\epsilon_0}\,\mathrm{d}r =$$
$$= \left[-\frac{ka}{r\epsilon_0}\right]_{b}^{a} - \left[\frac{k}{\epsilon_0}\ln(r)\right]_{b}^{a} = -\frac{k}{\epsilon_0} + \frac{ka}{b\epsilon_0} + \frac{k}{\epsilon_0}\ln\left(\frac{b}{a}\right)$$

and the potential difference from a to 0 is zero. Adding up the potential difference from $\infty \to b, b \to a$ gives

$$\frac{k}{\epsilon_0} - \frac{ka}{b\epsilon_0} - \frac{k}{\epsilon_0} + \frac{ka}{b\epsilon_0} + \frac{k}{\epsilon_0}\ln\left(\frac{b}{a}\right)$$
$$\downarrow$$
$$\boxed{V(0) = \frac{k}{\epsilon_0}\ln\left(\frac{b}{a}\right)}$$

□

Problem 2.25

Problem 2.26

Problem 2.27

Problem 2.28

Problem 2.29

Problem 2.30

Problem 2.31

(a) Check that the results of Exs. 2.5 and 2.6, and Prob. 2.12, are consistent with

$$\mathbf{E}_{\text{above}} - \mathbf{E}_{\text{below}} = \frac{\sigma}{\epsilon_0}\hat{\mathbf{n}} \tag{1}$$

(b) Use Gauss' law to find the field inside and outside a long hollow cylindrical tube, which carries a uniform surface charge σ. Check that your result is consistent with Eq. 1.

(c) Check that the result of Ex. 2.8 is consistent with boundary conditions

$$V_{\text{above}} = V_{\text{below}}$$
$$\frac{\partial V_{\text{above}}}{\partial n} - \frac{\partial V_{\text{below}}}{\partial n} = -\frac{1}{\epsilon_0}\sigma$$

Ex. 2.5 result

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0}\hat{\mathbf{n}}$$

and dividing it up between field above and field below

$$\mathbf{E}_{\text{above}} = \frac{\sigma}{\epsilon_0}\hat{\mathbf{n}}$$
$$\mathbf{E}_{\text{below}} = -\frac{\sigma}{\epsilon_0}\hat{\mathbf{n}}$$

gives

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0}\hat{\mathbf{n}}$$

Problem 2.32

(a) Three charges are situated at the corners of a square (side a), as shown in the figure. How much work does it take to bring in another charge, $+q$, from far away and place it in the fourth corner?

(b) How much work does it take to assemble the whole configuration of four charges?

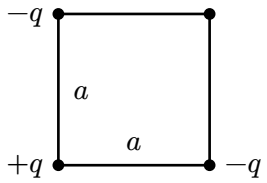


Figure 6:

Solution:

(a) The work required to bring in a charge q_4 is

$$W_4 = \frac{1}{4\pi\epsilon_0} q_4 \left(\frac{q_1}{r_{14}} + \frac{q_2}{r_{24}} + \frac{q_3}{r_{34}} \right)$$

where each part is due to the three charges, q_1, q_2, q_3, already in the configuration. The separation distances r are $r_{14} = a, r_{24} = \sqrt{2}a, r_{34} = a$

This gives

$$W_4 = \frac{1}{4\pi\epsilon_0} q^2 \left(-\frac{1}{a} + \frac{1}{\sqrt{2}a} - \frac{1}{a} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(-2 + \frac{1}{\sqrt{2}} \right)$$

(b)

$$W_1 = 0$$

$$W_2 = \frac{1}{4\pi\epsilon_0} q_2 \left(\frac{q_1}{r_{12}} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} (-1)$$

$$W_3 = \frac{1}{4\pi\epsilon_0} q_3 \left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(-1 + \frac{1}{\sqrt{2}} \right)$$

$$W_4 = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(-2 + \frac{1}{\sqrt{2}} \right)$$

$$W_1 + W_2 + W_3 + W_4 = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(-1 - 1 + \frac{1}{\sqrt{2}} - 2 + \frac{1}{\sqrt{2}} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(-4 + \sqrt{2} \right)$$

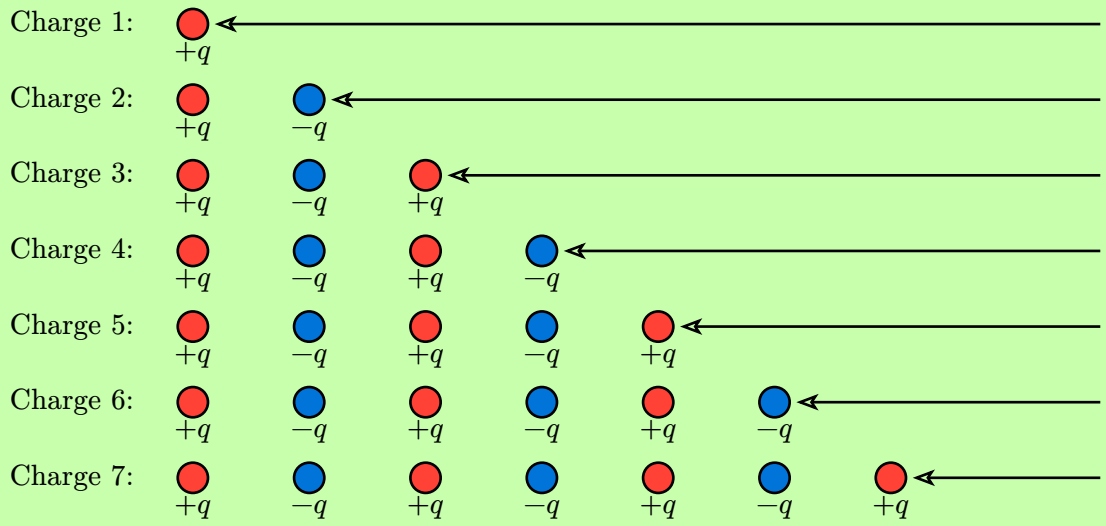
□

Problem 2.33

Problem 2.34

Consider an infinite chain of point charges, $\pm q$ (with alternating signs), strung out along the x axis, each a distance a from its nearest neighbors. Find the work per particle required to assemble this system.

Solution:



Adding charges 1-7 requires the work W_1-W_7:

$$
\begin{aligned}
W_1 &= 0 \\
W_2 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}(-1) \\
W_3 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}\left(-1+\frac{1}{2}\right) \\
W_4 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}\left(-1+\frac{1}{2}-\frac{1}{3}\right) \\
W_5 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}\left(-1+\frac{1}{2}-\frac{1}{3}+\frac{1}{4}\right) \\
W_6 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}\left(-1+\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}\right) \\
W_6 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}\left(-1+\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}+\frac{1}{6}\right) \\
W_7 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}\left(-1+\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}+\frac{1}{6}-\frac{1}{7}\right)
\end{aligned}
$$

As we keep adding charges, we add the constant $(-1)^n\frac{1}{n}$ which means the Madelung constant is

$$\alpha = \sum_1^\infty \frac{(-1)^n}{n}$$

which is similar to the known Maclaurin series for $\ln(x+1)$:

$$\ln(x+1) = \sum_1^\infty \frac{(-1)^{n+1}}{n}x^n = -\sum_1^\infty \frac{(-1)^n}{n}x^n$$

Setting $x = 1$ and moving the negative sign to the LHS, we get

$$-\ln(2) = \sum_1^\infty \frac{(-1)^n}{n}$$

which is identical to the Madelung constant, which gives

$$\boxed{\alpha = -\ln(2)}$$

□

Problem 2.35

$$W = \frac{1}{2}\int \rho V \, \mathrm{d}\tau \tag{2}$$

$$W = \frac{\epsilon_0}{2}\left(\int_{\mathcal{V}} E^2\,\mathrm{d}\tau + \oint_{\mathcal{S}} V\mathbf{E}\cdot\mathrm{d}\mathbf{a}\right) \tag{3}$$

$$W = \frac{\epsilon_0}{2}\int E^2\,\mathrm{d}\tau \tag{4}$$

Find the energy stored in a uniformly charged solid sphere of radius R and charge q. Do it three different ways:

(a) Use Eq. 2 You found the potential in Prob. 2.22.

(b) Use Eq. 4. Don't forget to integrate over all space.

(c) Use Eq. 3. Take a spherical volume of radius a. What happens as $a \to \infty$?

Solution:

(a) Using Eq. 2 with

$$V = \frac{q}{8\pi\epsilon_0 R}\left(3 - \frac{r^2}{R^2}\right)$$

gives the energy

$$W = \frac{1}{2}\int \frac{\rho q}{8\pi\epsilon_0 R}\left(3r^2 - \frac{r^4}{R^2}\right)\mathrm{d}\tau =$$

$$= \frac{1}{2}\frac{\rho q}{2\epsilon_0 R}\int_0^R \left(3r^2 - \frac{r^4}{R^2}\right)\mathrm{d}r = \frac{\rho q}{4\epsilon_0 R}\left[r^3 - \frac{r^5}{5R^2}\right]_0^R =$$

$$= \frac{\rho q}{4\epsilon_0 R}\left[R^3 - \frac{R^3}{5}\right] = \frac{\rho q}{4}\epsilon_0\left[R^2 - \frac{R^2}{5}\right] = \frac{\rho q}{5\epsilon_0}R^2$$

and with

$$\rho = \frac{q}{\frac{4}{3}\pi R^3}$$

we can simplify the expression to

$$W = \frac{3q^2}{20\pi\epsilon_0 R} = \frac{1}{4\pi\epsilon_0}\frac{3q^2}{5R}$$

(b) The electric field

$$\mathbf{E}_{\mathrm{o}} = \frac{q}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}}$$

$$\mathbf{E}_{\mathrm{i}} = \frac{qr}{4\pi\epsilon_0 R^3}\hat{\mathbf{r}}$$

Starting with the outside, E^2 becomes $\frac{q^2}{16\pi^2\epsilon_0^2 r^4}$ giving the energy

$$W = \frac{\epsilon_0}{2}\int \frac{q^2}{16\pi^2\epsilon_0^2 r^4}\,\mathrm{d}\tau = \frac{q^2\epsilon_0 4\pi}{32\pi^2\epsilon_0^2}\int_R^\infty \frac{1}{r^2} = \frac{q^2\epsilon_0 4\pi}{32\pi^2\epsilon_0^2}\left[-\frac{1}{r}\right]_R^\infty =$$

$$= \frac{q^2}{8\pi\epsilon_0 R}$$

Doing the same for the inside, E^2 becomes $\frac{q^2 r^2}{16\pi^2\epsilon_0^2 R^6}$ giving the energy

$$W = \frac{\epsilon_0}{2}\int \frac{q^2 r^2}{16\pi^2\epsilon_0^2 R^6}\,\mathrm{d}\tau = \frac{4\pi\epsilon_0 q^2}{32\pi^2\epsilon_0^2 R^6}\int_0^R r^4 =$$

$$= \frac{q^2}{8\pi\epsilon_0}\left(\frac{1}{5R}\right)$$

Adding them up gives

$$\boxed{W = \frac{1}{4\pi\epsilon_0}q^2\left(\frac{1}{2R}+\frac{1}{10R}\right) = \frac{1}{4\pi\epsilon_0}\frac{3q^2}{5R}}$$

(c) Using Eq. 3 we have the fields

$$E_i^2 = \frac{q^2 r^2}{16\pi^2\epsilon_0^2 R^6}$$

$$E_o^2 = \frac{q^2}{16\pi^2\epsilon_0^2 r^4}$$

$$V_o = \frac{q}{4\pi\epsilon_0 r}$$

This gives the energy

$$W = \frac{\epsilon_0}{2}\left(\frac{4\pi q^2}{16\pi^2\epsilon_0^2 R^6}\int_0^R r^4\,\mathrm{d}r + \frac{4\pi q^2}{16\pi^2\epsilon_0^2}\int_R^a \frac{1}{r^2}\,\mathrm{d}r + \frac{q^2}{16\pi^2\epsilon_0^2}\oint \frac{1}{r}\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi\right)$$

$$\downarrow$$

$$\frac{q^2}{2}\left(\frac{1}{4\pi\epsilon_0 R^6}\int_0^R r^4\,\mathrm{d}r + \frac{1}{4\pi\epsilon_0}\int_R^a \frac{1}{r^2}\,\mathrm{d}r + \frac{1}{4\pi\epsilon_0}\frac{1}{a}\right)$$

where the volume integrals were immediately simplified according to $\int_0^R\int_0^\pi\int_0^{2\pi}\mathrm{d}r\,\mathrm{d}\phi\,\mathrm{d}\theta = 4\pi\int_0^R \mathrm{d}r$

$$\frac{q^2}{2}\left(\frac{1}{4\pi\epsilon_0 R}\frac{1}{5} - \frac{1}{4\pi\epsilon_0 a} + \frac{1}{4\pi\epsilon_0}\frac{1}{R} + \frac{1}{4\pi\epsilon_0}\frac{1}{a}\right)$$

$$\downarrow$$

$$\boxed{W = \frac{1}{4\pi\epsilon_0}\frac{3q^2}{5R}}$$

□

Problem 2.36

Problem 2.37

Problem 2.38

Problem 2.39

Problem 2.40

Problem 2.41

Problem 2.42

Problem 2.43

Problem 2.44

Problem 2.45

Problem 2.46

Problem 2.47

Problem 2.48

Problem 2.49

Problem 2.50

Problem 2.51

Problem 2.52

Problem 2.53

Problem 2.54

Problem 2.55

Problem 2.56

Problem 2.57

Problem 2.58

Problem 2.59

Problem 2.60

Problem 2.61

Problem 2.62

Problem 2.63

Problem 2.64

Problem 2.65

Problem 2.66

Problem 2.67